

類別資料分析 第5章

The Analysis of Categorical Data

Chapter 5 : Building and Applying Logistic Regression Models

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Linear Regression Models

$y_i = \beta_0 + \beta_1 x_{1i} + \cdots + \beta_{p-1} x_{(p-1)i} + \varepsilon_i$ Multiple regression

$$\begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix}_{n \times 1} = \begin{pmatrix} 1 & x_{11} & \cdots & x_{(p-1),1} \\ 1 & x_{12} & \cdots & x_{(p-1),2} \\ \vdots & \vdots & \ddots & \vdots \\ 1 & x_{1n} & \cdots & x_{(p-1),n} \end{pmatrix}_{n \times p} \begin{pmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_{p-1} \end{pmatrix}_{p \times 1} + \begin{pmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \vdots \\ \varepsilon_n \end{pmatrix}_{n \times 1}$$

$$Y = X\beta + \varepsilon$$

$$\min_{\beta} \sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_{1i} - \cdots - \beta_{p-1} x_{(p-1)i})^2 \equiv \min_{\beta} (Y - X\beta)'(Y - X\beta) : \text{SSE value}$$

$$\frac{\partial Q(\beta)}{\partial \beta} = \begin{pmatrix} \frac{\partial Q(\beta)}{\partial \beta_0} \\ \vdots \\ \frac{\partial Q(\beta)}{\partial \beta_{p-1}} \end{pmatrix}_{p \times 1} = X'Y - X'X\beta = 0 \Rightarrow \hat{\beta} = (X'X)^{-1}X'Y \quad (Y - X\beta)'(Y - X\beta) = Q(\beta)_{1 \times 1}$$

$$\Rightarrow \hat{Y} = X\hat{\beta} = X(X'X)^{-1}X'Y = P_X Y : \text{fitted value}$$

$$X(X'X)^{-1}X' = P_X : \text{Projection matrix}$$

Linear Regression Models

$$Y - \hat{Y} = (I - P_X)Y$$
$$\because (I - P_X)' = (I - P_X) \quad \therefore (I - P_X) \text{ is a symmetric matrix}$$
$$(I - P_X)(I - P_X)' = I^2 - P_X - P_X + P_X^2 = I - P_X \quad \therefore (I - P_X) \text{ is idempotent.}$$
$$P_X^2 = X(X'X)^{-1}X' \cdot X(X'X)^{-1}X' = X(X'X)^{-1}X' = P_X \quad \therefore P_X \text{ is also idempotent.}$$
$$\Rightarrow \text{Var}(Y - \hat{Y}) = (I - P_X) \cdot \text{Var}(Y) \cdot (I - P_X)' = (I - P_X)\sigma^2$$

Logistic Regression

※類別資料分析和迴歸分析的比較

	類別資料分析	迴歸分析
資料型態	反應變數是離散型	反應變數是連續型
參數估計	MLE	LSE
假設檢定	LRT	t test or F test

Logistic Regression Models

※Dummy variables in logit models

(1) For the $2 \times 2 \times 2$ contingency table

Y : Binary response variable

X, Z : Binary explanatory variables

$$\pi = \pi(x, z) = P(Y = 1 | X = x, Z = z)$$

$$Y = \begin{cases} 1 & \text{success} \\ 0 & \text{failure} \end{cases}$$

$$X = \begin{cases} 1 & \text{drug A} \\ 0 & \text{drug B} \end{cases}$$

$$Z = \begin{cases} 1 & \text{clinic 1} \\ 0 & \text{clinic 2} \end{cases}$$

Y	Z	1	1	0	0
	X	1	0	1	0
0		π_0	π_1	π_2	π_3
1		$1 - \pi_0$	$1 - \pi_1$	$1 - \pi_2$	$1 - \pi_3$

Building and Applying Logistic Regression Models

※Dummy variables in logit models

(1) For the $2 \times 2 \times 2$ contingency table

· model : $\text{logit } \pi = \log \frac{\pi}{1-\pi} = \alpha + \beta_1 x + \beta_2 z$

Fixed $Z = 1$:

$$X = \begin{cases} 1 & \text{logit}(\pi_0) = \alpha + \beta_1 + \beta_2 \\ 0 & \text{logit}(\pi_1) = \alpha + \beta_2 \end{cases}$$

$$\therefore \text{logit}(\pi_0) - \text{logit}(\pi_1) = \log \frac{\pi_0/(1-\pi_0)}{\pi_1/(1-\pi_1)} = \beta_1$$

Y	Z	1	1	0	0
	X	1	0	1	0
0		π_0	π_1	π_2	π_3
1		$1 - \pi_0$	$1 - \pi_1$	$1 - \pi_2$	$1 - \pi_3$

$$\theta_{XY(1)} = \frac{\pi_0/(1-\pi_0)}{\pi_1/(1-\pi_1)} : \text{odds ratio of } Z = 1, X = 1 \text{ vs. } Z = 1, X = 0$$

Building and Applying Logistic Regression Models

※Dummy variables in logit models

(1) For the $2 \times 2 \times 2$ contingency table

· model : $\text{logit } \pi = \log \frac{\pi}{1-\pi} = \alpha + \beta_1 X + \beta_2 Z$

Fixed $Z = 0$:

$$X = \begin{cases} 1 & \text{logit}(\pi_2) = \alpha + \beta_1 \\ 0 & \text{logit}(\pi_3) = \alpha \end{cases}$$

$$\theta_{XY(0)} = \frac{\pi_2/(1-\pi_2)}{\pi_3/(1-\pi_3)} : \text{odds ratio of } Z = 0, X = 1 \text{ vs. } Z = 0, X = 0$$

$$\therefore \text{logit}(\pi_2) - \text{logit}(\pi_3) = \log \frac{\pi_2/(1-\pi_2)}{\pi_3/(1-\pi_3)} = \beta_1$$

$\Rightarrow$ 不管 $Z = 0$ 或 $Z = 1$ · X & Y 的 conditional odds ratio 相同 · 且 $\theta_{XY(1)} = \theta_{XY(0)} = e^{\beta_1}$

Y	Z	1	1	0	0
	X	1	0	1	0
	0	π_0	π_1	π_2	π_3
	1	$1 - \pi_0$	$1 - \pi_1$	$1 - \pi_2$	$1 - \pi_3$

Building and Applying Logistic Regression Models

※Dummy variables in logit models

(1) For the $2 \times 2 \times 2$ contingency table

· model : $\text{logit } \pi = \log \frac{\pi}{1-\pi} = \alpha + \beta_1 X + \beta_2 Z + \beta_3 XZ$ (saturated model)

Fixed $Z = 1$:

$$X = \begin{cases} 1 & \text{logit}(\pi_0) = \alpha + \beta_1 + \beta_2 + \beta_3 \\ 0 & \text{logit}(\pi_1) = \alpha + \beta_2 \end{cases}$$

$$\therefore \text{logit}(\pi_0) - \text{logit}(\pi_1) = \log \frac{\pi_0/(1-\pi_0)}{\pi_1/(1-\pi_1)} = \beta_1 + \beta_3$$

Y	Z	1	1	0	0
	X	1	0	1	0
	0	π_0	π_1	π_2	π_3
	1	$1 - \pi_0$	$1 - \pi_1$	$1 - \pi_2$	$1 - \pi_3$

Building and Applying Logistic Regression Models

※Dummy variables in logit models

(1) For the $2 \times 2 \times 2$ contingency table

· model : $\text{logit } \pi = \log \frac{\pi}{1-\pi} = \alpha + \beta_1 x + \beta_2 z + \beta_3 xz$ (saturated model)

Fixed $Z = 0$:

$$X = \begin{cases} 1 & \text{logit}(\pi_2) = \alpha + \beta_1 \\ 0 & \text{logit}(\pi_3) = \alpha \end{cases}$$

$$\therefore \text{logit}(\pi_2) - \text{logit}(\pi_3) = \log \frac{\pi_2/(1-\pi_2)}{\pi_3/(1-\pi_3)} = \beta_1$$

⇒ 當 $Z = 0$ · $X \& Y$ 的 conditional odds ratio 為 $\theta_{XY(1)} = e^{\beta_1 + \beta_3}$

當 $Z = 1$ · $X \& Y$ 的 conditional odds ratio 為 $\theta_{XY(0)} = e^{\beta_1}$

Y	Z	1	1	0	0
	X	1	0	1	0
	0	π_0	π_1	π_2	π_3
	1	$1 - \pi_0$	$1 - \pi_1$	$1 - \pi_2$	$1 - \pi_3$

Building and Applying Logistic Regression Models

※Dummy variables in logit models

(1) For the $2 \times 2 \times 2$ contingency table

· model : $\text{logit } \pi = \log \frac{\pi}{1-\pi} = \alpha + \beta_1 x + \beta_2 z + \beta_3 xz$ (saturated model)

$$\begin{cases} H_0 : \beta_3 = 0 \\ H_1 : \text{logit}(\pi) = \alpha + \beta_1 x + \beta_2 z + \beta_3 xz \end{cases} \equiv \begin{cases} H_0 : \theta_{XY(1)} = \theta_{XY(0)} \\ H_0 : \theta_{XY(1)} \neq \theta_{XY(0)} \end{cases} : \text{Breslow-Day homogeneity test}$$

$$\begin{cases} H_0 : \beta_1 = 0 \\ H_1 : \text{logit}(\pi) = \alpha + \beta_1 x + \beta_2 z \end{cases} \equiv \begin{cases} H_0 : \theta_{XY(1)} = \theta_{XY(0)} = 1 \\ H_0 : \text{at least one } \theta_{XY(i)} \neq 1 \end{cases} : \text{Mantel-Haenszel test}$$

Building and Applying Logistic Regression Models

※Dummy variables in logit models

(1) For the $2 \times 2 \times 2$ contingency table

$$\text{若令 } X^* = \begin{cases} 0 & \text{drug A} \\ 1 & \text{drug B} \end{cases} \quad Z^* = \begin{cases} 0 & \text{clinic 1} \\ 1 & \text{clinic 2} \end{cases}$$

· model : $\text{logit } \pi = \alpha^* + \beta_1^* X^* + \beta_2^* Z^*$

Fixed $Z^* = 0$:

$$\begin{cases} X^* = 0 & \text{logit}(\pi_0) = \alpha^* = \alpha + \beta_1 + \beta_2 \\ X^* = 1 & \text{logit}(\pi_1) = \alpha^* + \beta_1^* = \alpha + \beta_2 \end{cases}$$

$$\therefore \text{logit}(\pi_0) - \text{logit}(\pi_1) = \log \frac{\pi_0/(1-\pi_0)}{\pi_1/(1-\pi_1)} = -\beta_1^*$$

$$\theta_{XY(0)} = e^{-\beta_1^*}$$

Building and Applying Logistic Regression Models

※Dummy variables in logit models

(1) For the $2 \times 2 \times 2$ contingency table

· model : $\text{logit } \pi = \alpha^* + \beta_1^* X^* + \beta_2^* Z^*$

Fixed $Z^* = 1$:

$$\begin{cases} X^* = 0 & \text{logit}(\pi_2) = \alpha^* + \beta_2^* = \alpha + \beta_1 \\ X^* = 1 & \text{logit}(\pi_3) = \alpha^* + \beta_1^* + \beta_2^* = \alpha \end{cases}$$

$$\therefore \text{logit}(\pi_2) - \text{logit}(\pi_3) = \log \frac{\pi_2/(1-\pi_2)}{\pi_3/(1-\pi_3)} = -\beta_1^*$$

$$\theta_{XY(1)} = e^{-\beta_1^*}$$

$$\Rightarrow \alpha^* = \alpha + \beta_1 + \beta_2, \beta_1^* = -\beta_1, \beta_2^* = -\beta_2$$

Building and Applying Logistic Regression Models

※Dummy variables in logit models

(1) For the $2 \times 2 \times 2$ contingency table

H. W. :若 model為 $\text{logit } \pi = \alpha^* + \beta_1^* x^* + \beta_2^* z^* + \beta_3^* x^* z^*$

其中 $X^* = \begin{cases} 0 & \text{drug A} \\ 1 & \text{drug B} \end{cases}$ $Z^* = \begin{cases} 0 & \text{clinic 1} \\ 1 & \text{clinic 2} \end{cases}$

試求 $(\alpha^*, \beta_1^*, \beta_2^*, \beta_3^*)$ 與 $(\alpha, \beta_1, \beta_2, \beta_3)$ 的關係?

Building and Applying Logistic Regression Models

※Dummy variables in logit models

(2) For the 2×3 contingency table

Y : Binary response variable

X, Z : Explanatory variables of 3 levels

Let (X^*, Z^*) be the dummy variables of X

$\pi = \pi(x, z) = P(Y = 1|X = x) = P(Y = 1|X^* = x^*, Z^* = z^*)$

Y	X	0	1	2
	X*	0	1	0
	Z*	0	0	1
	0	π_0	π_1	π_2
	1	$1 - \pi_0$	$1 - \pi_1$	$1 - \pi_2$

Building and Applying Logistic Regression Models

※Dummy variables in logit models

(2) For the 2×3 contingency table

· model 1 : $\text{logit } \pi = \log \frac{\pi}{1-\pi} = \alpha^* + \beta_1^* x^* + \beta_2^* z^*$

$X = 0 \Rightarrow (X^*, Z^*) = (0, 0) \Rightarrow \text{logit}(\pi_0) = \alpha^*$

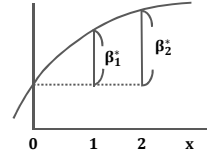
$X = 1 \Rightarrow (X^*, Z^*) = (1, 0) \Rightarrow \text{logit}(\pi_1) = \alpha^* + \beta_1^*$

$X = 2 \Rightarrow (X^*, Z^*) = (0, 1) \Rightarrow \text{logit}(\pi_2) = \alpha^* + \beta_2^*$

$\therefore \text{logit}(\pi_2) - \text{logit}(\pi_0) = \log \frac{\pi_2/(1-\pi_2)}{\pi_0/(1-\pi_0)} = \beta_2^*$

$\text{logit}(\pi_1) - \text{logit}(\pi_0) = \log \frac{\pi_1/(1-\pi_1)}{\pi_0/(1-\pi_0)} = \beta_1^*$

· 當 $\beta_2^* = 2 \cdot \beta_1^*$ · 則 model 1 就 reduce 為 linear model (model 2)



- $X = 2$ vs. $X = 0$ 的 odds ratio 為 $e^{\beta_2^*}$
- $X = 1$ vs. $X = 0$ 的 odds ratio 為 $e^{\beta_1^*}$
- $X = 2$ vs. $X = 1$ 的 odds ratio 為 $e^{\beta_2^* - \beta_1^*}$

Building and Applying Logistic Regression Models

※Dummy variables in logit models

(2) For the 2×3 contingency table

· model 2 : $\text{logit } \pi = \log \frac{\pi}{1-\pi} = \alpha + \beta_1 x$

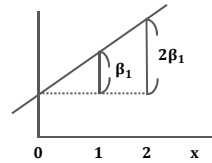
$X = 0 \Rightarrow \text{logit}(\pi_0) = \alpha$

$X = 1 \Rightarrow \text{logit}(\pi_1) = \alpha + \beta_1$

$X = 2 \Rightarrow \text{logit}(\pi_2) = \alpha + 2\beta_1$

$\therefore \text{logit}(\pi_2) - \text{logit}(\pi_0) = \log \frac{\pi_2/(1-\pi_2)}{\pi_0/(1-\pi_0)} = 2\beta_1$

$\text{logit}(\pi_1) - \text{logit}(\pi_0) = \log \frac{\pi_1/(1-\pi_1)}{\pi_0/(1-\pi_0)} = \beta_1$



- $X = 2$ vs. $X = 0$ 的 odds ratio 為 $e^{2\beta_1}$
- $X = 1$ vs. $X = 0$ 的 odds ratio 為 e^{β_1}
- $X = 2$ vs. $X = 1$ 的 odds ratio 為 $e^{2\beta_1 - \beta_1} = e^{\beta_1}$

Building and Applying Logistic Regression Models

※Dummy variables in logit models

(3) For the $2 \times 2 \times k$ contingency table

Y : Binary response variable

X : Explanatory variables of 2 levels

Z : Explanatory variables of k levels

$\pi = \pi(x, z) = P(Y = 1 | X = x, Z = z)$

Z = 1		Y	
		1	0
X	1	π_1	$1 - \pi_1$
	0	π_2	$1 - \pi_2$

$$\theta_{XY(1)} = \frac{\pi_1 / (1 - \pi_1)}{\pi_2 / (1 - \pi_2)}$$

Z = 2		Y	
		1	0
X	1	π_3	$1 - \pi_3$
	0	π_4	$1 - \pi_4$

$$\theta_{XY(2)} = \frac{\pi_3 / (1 - \pi_3)}{\pi_4 / (1 - \pi_4)}$$

...

Z = k		Y	
		1	0
X	1	π_{2k-1}	$1 - \pi_{2k-1}$
	0	π_{2k}	$1 - \pi_{2k}$

$$\theta_{XY(k)} = \frac{\pi_{2k-1} / (1 - \pi_{2k-1})}{\pi_{2k} / (1 - \pi_{2k})}$$

Building and Applying Logistic Regression Models

※Dummy variables in logit models

(3) For the $2 \times 2 \times k$ contingency table

· model : $\text{logit } \pi = \log \frac{\pi}{1-\pi} = \alpha + \beta_1 x + \beta_2 z$

Fixed $X = 1$:

$Z = 1 \Rightarrow \text{logit}(\pi_1) = \alpha + \beta_1 + \beta_2$

$Z = 2 \Rightarrow \text{logit}(\pi_3) = \alpha + \beta_1 + 2\beta_2$

...

$Z = k \Rightarrow \text{logit}(\pi_{2k-1}) = \alpha + \beta_1 + k\beta_2$

X = 1		Y	
		1	0
Z	1	π_1	$1 - \pi_1$
	2	π_3	$1 - \pi_3$
	3	π_5	$1 - \pi_5$
	4	π_7	$1 - \pi_7$
	...	...	...
	k-1	π_{2k-3}	$1 - \pi_{2k-3}$
	k	π_{2k-1}	$1 - \pi_{2k-1}$

Building and Applying Logistic Regression Models

※Dummy variables in logit models

(3) For the $2 \times 2 \times k$ contingency table

$$\cdot \text{model : } \text{logit } \pi = \log \frac{\pi}{1-\pi} = \alpha + \beta_1 X + \beta_2 Z$$

Fixed $X = 1$:

$$\therefore \text{logit}(\pi_3) - \text{logit}(\pi_1) = \log \frac{\pi_3/(1-\pi_3)}{\pi_1/(1-\pi_1)} = \beta_2$$

$$\text{logit}(\pi_5) - \text{logit}(\pi_1) = \log \frac{\pi_5/(1-\pi_5)}{\pi_1/(1-\pi_1)} = 2\beta_2$$

$$\text{logit}(\pi_{2k-1}) - \text{logit}(\pi_1) = \log \frac{\pi_{2k-1}/(1-\pi_{2k-1})}{\pi_1/(1-\pi_1)} = (2k-1)\beta_2$$

X = 1		Y	
		1	0
Z	1	π_1	$1 - \pi_1$
	2	π_3	$1 - \pi_3$
	3	π_5	$1 - \pi_5$
	4	π_7	$1 - \pi_7$
	...	...	...
	k-1	π_{2k-3}	$1 - \pi_{2k-3}$
	k	π_{2k-1}	$1 - \pi_{2k-1}$

Building and Applying Logistic Regression Models

※Dummy variables in logit models

(3) For the $2 \times 2 \times k$ contingency table

$$\cdot \text{model : } \text{logit } \pi = \log \frac{\pi}{1-\pi} = \alpha + \beta_1 X + \beta_2 Z$$

Fixed $X = 1$:

∴ 不管 $X = 1$ 或 $X = 0$ · Z & Y 的 conditional odds ratio

$$Z = 2 \text{ vs. } Z = 1 \text{ 為 } \theta_{21} = e^{\beta_2}$$

$$Z = 3 \text{ vs. } Z = 1 \text{ 為 } \theta_{31} = e^{2\beta_2}$$

$$Z = k \text{ vs. } Z = 1 \text{ 為 } \theta_{k1} = e^{(k-1)\beta_2}$$

Building and Applying Logistic Regression Models

※Dummy variables in logit models

(3) For the $2 \times 2 \times k$ contingency table

· model : $\text{logit } \pi = \log \frac{\pi}{1-\pi} = \alpha + \beta_1 x + \beta_2 z$

Fixed $Z = 1$:

$$X = 1 \Rightarrow \text{logit}(\pi_1) = \alpha + \beta_1 + \beta_2$$

$$X = 0 \Rightarrow \text{logit}(\pi_2) = \alpha + \beta_2$$

$$\therefore \text{logit}(\pi_1) - \text{logit}(\pi_2) = \log \frac{\pi_1/(1-\pi_1)}{\pi_2/(1-\pi_2)} = \beta_1 \quad \theta_{XY(1)} = \frac{\pi_1/(1-\pi_1)}{\pi_2/(1-\pi_2)}$$

$\Rightarrow$ 不管 Z 為何值 · X & Y 的 conditional odds ratio 皆相同 : $\theta_{XY(1)} = \theta_{XY(2)} = \dots = \theta_{XY(k)} = e^{\beta_1}$

Building and Applying Logistic Regression Models

※Dummy variables in logit models

(3) For the $2 \times 2 \times k$ contingency table

將 Z 設為 Dummy variables : $Z_1^*, Z_2^*, \dots, Z_{k-1}^*$

· model : $\text{logit } \pi = \log \frac{\pi}{1-\pi} = \alpha^* + \beta_1^* x + \beta_2^* z_1^* + \dots + \beta_k^* z_{k-1}^*$

$$Z_1^* = \begin{cases} 1 & \text{if } Z = 2 \\ 0 & \text{otherwise} \end{cases}$$

$$Z_2^* = \begin{cases} 1 & \text{if } Z = 3 \\ 0 & \text{otherwise} \end{cases} \Rightarrow$$

$$Z_{k-1}^* = \begin{cases} 1 & \text{if } Z = k \\ 0 & \text{otherwise} \end{cases}$$

Z	$(Z_1^*, Z_2^*, \dots, Z_{k-1}^*)$
$Z = 1$	$(0, 0, \dots, 0)$
$Z = 2$	$(1, 0, \dots, 0)$
$Z = 3$	$(0, 1, \dots, 0)$
$\dots$	$\dots$
$Z = k$	$(0, 0, \dots, 1)$

Building and Applying Logistic Regression Models

※Dummy variables in logit models

(3) For the $2 \times 2 \times k$ contingency table

$$\cdot \text{ model : } \text{logit } \pi = \log \frac{\pi}{1-\pi} = \alpha^* + \beta_1^* x + \beta_2^* z_1 + \cdots + \beta_k^* z_{k-1}$$

Fixed $X = 1$, Z using dummy variables :

$$Z = 1 \Rightarrow \text{logit}(\pi_1) = \alpha^* + \beta_1^*$$

$$Z = 2 \Rightarrow \text{logit}(\pi_3) = \alpha^* + \beta_1^* + \beta_2^*$$

...

$$Z = k \Rightarrow \text{logit}(\pi_{2k-1}) = \alpha^* + \beta_1^* + \beta_k^*$$

Building and Applying Logistic Regression Models

※Dummy variables in logit models

(3) For the $2 \times 2 \times k$ contingency table

$$\cdot \text{ model : } \text{logit } \pi = \log \frac{\pi}{1-\pi} = \alpha^* + \beta_1^* x + \beta_2^* z_1 + \cdots + \beta_k^* z_{k-1}$$

Fixed $X = 1$:

$$\therefore \text{logit}(\pi_3) - \text{logit}(\pi_1) = \log \frac{\pi_3/(1-\pi_3)}{\pi_1/(1-\pi_1)} = \beta_2^*$$

$$\text{logit}(\pi_5) - \text{logit}(\pi_1) = \log \frac{\pi_5/(1-\pi_5)}{\pi_1/(1-\pi_1)} = \beta_3^*$$

$$\text{logit}(\pi_{2k-1}) - \text{logit}(\pi_1) = \log \frac{\pi_{2k-1}/(1-\pi_{2k-1})}{\pi_1/(1-\pi_1)} = \beta_k^*$$

Building and Applying Logistic Regression Models

※Dummy variables in logit models

(3) For the $2 \times 2 \times k$ contingency table

$$\cdot \text{model : } \text{logit } \pi = \log \frac{\pi}{1-\pi} = \alpha^* + \beta_1^* x + \beta_2^* z_1 + \cdots + \beta_k^* z_{k-1}$$

Fixed $X = 1$:

$\therefore$ 不管 $X = 1$ 或 $X = 0$ · $Z - Y$ 的 conditional odds ratio

$$Z = 2 \text{ vs. } Z = 1 \text{ 為 } \theta_{21} = e^{\beta_2^*}$$

$$Z = 3 \text{ vs. } Z = 1 \text{ 為 } \theta_{31} = e^{\beta_3^*}$$

...

$$Z = k \text{ vs. } Z = 1 \text{ 為 } \theta_{k1} = e^{\beta_k^*}$$

Building and Applying Logistic Regression Models

※Dummy variables in logit models

(3) For the $2 \times 2 \times k$ contingency table

$$\cdot \text{model : } \text{logit } \pi = \log \frac{\pi}{1-\pi} = \alpha^* + \beta_1^* x + \beta_2^* z_1 + \cdots + \beta_k^* z_{k-1}$$

Fixed $Z = 1$:

$$X = 1 \Rightarrow \text{logit}(\pi_1) = \alpha^* + \beta_1^*$$

$$X = 0 \Rightarrow \text{logit}(\pi_2) = \alpha^*$$

$$\therefore \text{logit}(\pi_1) - \text{logit}(\pi_2) = \log \frac{\pi_1/(1-\pi_1)}{\pi_2/(1-\pi_2)} = \beta_1^*$$

$\Rightarrow$ 不管 Z 為何值 · X & Y 的 conditional odds ratio 皆為 $e^{\beta_1^*}$

Building and Applying Logistic Regression Models

※Dummy variables in logit models

(3) For the $2 \times 2 \times k$ contingency table

H. W. 若將 Z 設為 Dummy variables : $Z_1^*, Z_2^*, \dots, Z_{k-1}^*$

$$\text{model : logit } \pi = \log \frac{\pi}{1-\pi} = \alpha^* + \beta_1^* x + \beta_2^* Z_1^* + \dots + \beta_k^* Z_{k-1}^*$$

$$Z_1^* = \begin{cases} 1 & \text{if } Z = 2 \\ 0 & \text{otherwise} \\ -1 & \text{if } Z = k \end{cases} \quad Z_2^* = \begin{cases} 1 & \text{if } Z = 3 \\ 0 & \text{otherwise} \\ -1 & \text{if } Z = k \end{cases} \Rightarrow$$

...

$$Z_{k-1}^* = \begin{cases} 1 & \text{if } Z = k-1 \\ 0 & \text{otherwise} \\ -1 & \text{if } Z = k \end{cases}$$

Z	($Z_1^*, Z_1^*, \dots, Z_{k-1}^*$)
Z = 1	(1, 0, 0, ..., 0)
Z = 2	(0, 1, 0, ..., 0)
Z = 3	(0, 0, 1, ..., 0)
...	...
Z = k	(-1, -1, ..., -1)

column sum = 0

Building and Applying Logistic Regression Models

※Dummy variables in logit models

(4) Multiple logistic regression

Y : Binary response variable

$X_1, \dots, X_k$: Explanatory variables can be quantitative or qualitative

$$\pi = \pi(x_1, \dots, x_k) = P(Y = 1 | X_1 = x_1, \dots, X_k = x_k)$$

Building and Applying Logistic Regression Models

※ Dummy variables in logit models

(4) Multiple logistic regression

· model : $\text{logit } \pi = \alpha + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_k x_k$

Fixed $X_j, j \neq i : (X_i : \text{continuous r. v.})$

$$X_i = r + 1 \Rightarrow \text{logit}(\pi_{r+1}) = \alpha + \beta_1 x_1 + \cdots + \beta_i \cdot (r + 1) + \cdots + \beta_k x_k$$

$$X_i = r \Rightarrow \text{logit}(\pi_r) = \alpha + \beta_1 x_1 + \cdots + \beta_i \cdot r + \cdots + \beta_k x_k$$

$$\therefore \text{logit}(\pi_{r+1}) - \text{logit}(\pi_r) = \log \frac{\pi_{r+1}/(1-\pi_{r+1})}{\pi_r/(1-\pi_r)} = \beta_i \Rightarrow \theta_{X_i Y} = \frac{\pi_{r+1}/(1-\pi_{r+1})}{\pi_r/(1-\pi_r)} = e^{\beta_i}$$

即固定其他解釋變數 $X_j, j \neq i$ ，當 X_i 增加 1 單位， X_i & Y 的 conditional odds ratio 為原來的 e^{β_i} 倍。
(X_i & Y 的 conditional odds ratio 增加 $e^{\beta_i} - 1$ 倍)

Building and Applying Logistic Regression Models

※ **Model selection** with several predictors (Stepwise Selection Procedure)

假設有 4 個解釋變數，首先 fit 下列模式：

$$(1) \text{logit } \pi = \alpha + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + \beta_4 x_4 \quad (1 / 2 / 3 / 4)$$

$$(2) \text{logit } \pi = \alpha + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + \beta_4 x_4 + \beta_5 x_1 x_2 + \beta_6 x_1 x_3 + \beta_7 x_1 x_4 + \beta_8 x_2 x_3 + \beta_9 x_2 x_4 \\ + \beta_{10} x_3 x_4 \quad (12 / 13 / 14 / 23 / 24 / 34)$$

$$(3) \text{logit } \pi = \alpha + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + \beta_4 x_4 + \beta_5 x_1 x_2 + \beta_6 x_1 x_3 + \beta_7 x_1 x_4 + \beta_8 x_2 x_3 + \beta_9 x_2 x_4 \\ + \beta_{10} x_3 x_4 + \beta_{11} x_1 x_2 x_3 + \beta_{12} x_1 x_2 x_4 + \beta_{13} x_1 x_3 x_4 + \beta_{14} x_2 x_3 x_4 \quad (123 / 124 / 134 / 234)$$

Building and Applying Logistic Regression Models

※ **Model selection** with several predictors (Stepwise Selection Procedure)

可能有的情況：

- a. 若 model (3) 不適合，則用 Saturated model
- b. 若 model (3) 適合，model (2) 不適合，則可用 model (2) 開始作 Forward selection，
或用 model (3) 開始作 backward elimination
- c. 若 model (2) 適合，model (1) 不適合，則可用 model (1) 開始作 Forward selection，
或用 model (2) 開始作 backward elimination

Building and Applying Logistic Regression Models

※ **Model selection with several predictors (Stepwise Selection Procedure)**

假設是 c 的情況：

· **Forward selection：**

(I) 先作 6 個 G^2 檢定：
$$\begin{cases} H_0 : \text{model (1)} \\ H_1 : \text{model (1)} + \beta x_i x_j \end{cases}$$

看所有 significant 檢定中，(i. e. $p - \text{value} < \alpha$)，哪一個 $p - \text{value}$ 最小，則加入該

interaction，並設此 model 為 (a)，e. g. $\text{model (1)} + \beta_5 x_1 x_2$

Building and Applying Logistic Regression Models

※ Model selection with several predictors (Stepwise Selection Procedure)

假設是 c 的情況：

· **Forward selection :**

(II) 再作下列 G^2 檢定：
$$\begin{cases} H_0 : \text{model (a)} \\ H_1 : \text{model (a)} + \text{other } X_i X_j \end{cases}$$

看所有 significant 檢定中，(i.e. $p - \text{value} < \alpha$)，哪一個 $p - \text{value}$ 最小，則加入該

interaction，並設此 model 為 (b)，e. g. model (1) + $\beta_5 x_1 x_2$ + $\beta_6 x_1 x_3$

Building and Applying Logistic Regression Models

※ Model selection with several predictors (Stepwise Selection Procedure)

假設是 c 的情況：

· **Forward selection :**

(III) 再檢定是否可 Drop model (b) 其中之一的 interaction，e. g.

$$\begin{cases} H_0 : \text{model (1)} + \beta_6 x_1 x_3 \\ H_1 : \text{model (a)} + \beta_5 x_1 x_2 + \beta_6 x_1 x_3 \end{cases}$$

若該檢定不 significant，則去掉 $\beta_5 x_1 x_2$ ，否則，保留 $\beta_5 x_1 x_2$

(IV) 重複步驟 II 和 III，直到無法再加入或去掉 interaction。

Building and Applying Logistic Regression Models

※ Model selection with several predictors (Stepwise Selection Procedure)

假設是 c 的情況：

· **Backward elimination :**

(I) 先檢定 $\begin{cases} H_0 : \text{model (2)} - \beta_{x_i x_j} \\ H_1 : \text{model (2)} \end{cases}$

選擇 p - value 最大者，即最不顯著有差別者，將 model (2) 中之該項 interaction 去掉，

並設此 model 為 (a')，e. g. model (2) - $\beta_5 x_1 x_2$ (i. e. 13 / 14 / 23 / 24 / 34)

Building and Applying Logistic Regression Models

※ Model selection with several predictors (Stepwise Selection Procedure)

假設是 c 的情況：

· **Backward elimination :**

(II) 再檢定 $\begin{cases} H_0 : \text{model (a')} - \text{other } X_i X_j \\ H_1 : \text{model (a')} \end{cases}$

在所有不顯著之檢定當中，選擇 p - value 最大者，去掉該項 interaction，

並設此 model 為 (b')，e. g. model (2) - $\beta_5 x_1 x_2 - \beta_6 x_1 x_3$ (i. e. 14 / 23 / 24 / 34)

Building and Applying Logistic Regression Models

※ Model selection with several predictors (Stepwise Selection Procedure)

假設是 c 的情況：

· **Backward elimination**：

(III) 再檢定 model (a') 中丟掉之 interaction 是否可加回來改進 model，e. g.
$$\begin{cases} H_0 : 14/23/24/34 \\ H_1 : 12/14/23/24/34 \end{cases}$$

若顯著，則再加回 $\beta_5 x_1 x_2$ ，否則，保持 model (a')

(IV) 重複步驟 II 和 III，直到無法再加入或丟掉

Building and Applying Logistic Regression Models

※ Model selection with several predictors (Stepwise Selection Procedure)

· 為何類別資料分析不計算 R^2 ？(評估模型的好壞改為計算 $-2 \log \text{likelihood}$ ，AIC 或 SC)

由於在 logistic regression model，Y 為二元變數，Y 的值只是用以識別，

而 R^2 為引進自變數後，Y 之變異數被解釋掉之比例，

既然 Y 的值只是代號，因此 Y 的平均數和變異數會隨著 Y 給定的數值不同而改變，

R^2 也因此會隨著 Y 給定的數值不同而改變。

Building and Applying Logistic Regression Models

※Model selection with several predictors (Stepwise Selection Procedure)

$$\text{e. g. } \begin{cases} y_i = 1, i = 1, \dots, r \\ y_i = 0, i = r + 1, \dots, n \end{cases} \quad \pi_i = P(y_i = 1 | X_1 = x_{1i}, X_2 = x_{2i})$$

$$\text{model : logit } \pi = \alpha + \beta_1 x_1 + \beta_2 x_2$$

$$\Rightarrow \hat{y}_i = \hat{\pi}_i = \frac{e^{\hat{\alpha} + \hat{\beta}_1 x_1 + \hat{\beta}_2 x_2}}{1 + e^{\hat{\alpha} + \hat{\beta}_1 x_1 + \hat{\beta}_2 x_2}}$$

$$\therefore R^2 = \frac{SSR}{SST} = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2} = \text{Corr}(y_i, \hat{y}_i) \quad \text{當 } y_i = 1 \text{ 或 } y_i = 0$$

$$\text{其中 } \sum_{i=1}^n (y_i - \hat{y}_i)^2 = \sum_{i=1}^r (1 - \hat{\pi}_i)^2 + \sum_{i=r+1}^n (\hat{\pi}_i)^2 \quad \cdot \quad \sum_{i=1}^n (y_i - \bar{y})^2 = r(1 - \bar{y})^2 + (n - r)\bar{y}^2$$

$$\text{若 } \begin{cases} y_i = 1, i = 1, \dots, r \\ y_i = -1, i = r + 1, \dots, n \end{cases} \Rightarrow R^2 = ?$$

· 另外 · 當模式中包含較多解釋變數時 · R^2 不一定增加 · 而且 R^2 與解釋變數的值域有關

Building and Applying Logistic Regression Models

※Sample size for comparing two proportions

$$(1) \begin{cases} H_0 : \pi_1 = \pi_2 \\ H_1 : \pi_1 < \pi_2 \end{cases}$$

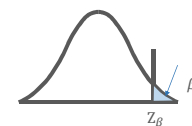
$$(2) Z = \frac{\hat{\pi}_1 - \hat{\pi}_2}{\sqrt{\frac{\pi_1(1-\pi_1)}{n_1} + \frac{\pi_2(1-\pi_2)}{n_2}}} \approx N(0, 1)$$

$$(3) \text{ Reject } H_0 \text{ if } |Z| > Z_{\alpha/2}$$

$$\begin{aligned} \Rightarrow \beta &= P\left(\hat{\pi}_1 - \hat{\pi}_2 > -Z_{\frac{\alpha}{2}} \sqrt{\frac{\pi_1(1-\pi_1) + \pi_2(1-\pi_2)}{n}} \mid H_1 \text{ is true}\right) \\ &= P\left(\frac{\hat{\pi}_1 - \hat{\pi}_2 - (\pi_1 - \pi_2)}{\sqrt{\frac{\pi_1(1-\pi_1) + \pi_2(1-\pi_2)}{n}}} > -Z_{\frac{\alpha}{2}} - \frac{(\pi_1 - \pi_2)}{\sqrt{\frac{\pi_1(1-\pi_1) + \pi_2(1-\pi_2)}{n}}} \mid H_1 \text{ is true}\right) \end{aligned}$$

$$\therefore n_1 = n_2 = n = \frac{(Z_{\alpha/2} + Z_{\beta})^2 [\pi_1(1-\pi_1) + \pi_2(1-\pi_2)]}{(\pi_1 - \pi_2)^2}$$

		Y		
		1	0	
X	1	π_1	$1 - \pi_1$	n_1
	0	π_2	$1 - \pi_2$	n_2



$$-Z_{\frac{\alpha}{2}} - \frac{(\pi_1 - \pi_2)}{\sqrt{\frac{\pi_1(1-\pi_1) + \pi_2(1-\pi_2)}{n}}} = Z_{\beta}$$

Building and Applying Logistic Regression Models

※ Sample size for comparing two proportions

e. g. $\pi_1 = 0.2$, $\pi_2 = 0.3$, $\alpha = 0.05$, $\beta = 0.10$, $Z_{\alpha/2} = 1.96$, $Z_\beta = 1.28$

$$\Rightarrow n_1 = n_2 = \frac{(1.96 + 1.28)^2 (0.2 \cdot 0.8 + 0.3 \cdot 0.7)}{(0.2 - 0.3)^2} = 389$$

Building and Applying Logistic Regression Models

※ Sample size with a quantitative predictor

[Hsieh, F. Y. (1989). Sample size tables for logistic regression. *Statistics in Medicine*, 8(7), 795-802.]

$$\begin{cases} H_0 : \beta_1 = 0 \\ H_1 : \text{logit } \pi = \beta_0 + \beta_1 x \end{cases}$$

$$X = \begin{cases} 1 & \text{group 1} \\ 0 & \text{group 2} \end{cases}$$

Let $\lambda = \log \theta$, $\theta = \frac{\pi(\bar{x})}{\pi(\bar{x} + S_X)}$, where $\bar{x}$ is the sample mean of X , S_X is the sd of X

$$\Rightarrow n = \frac{\left[Z_\alpha + Z_\beta \cdot \exp\left\{-\frac{\lambda^2}{4}\right\} \right]^2 (1 + 2 \cdot \bar{\pi} \cdot \delta)}{\bar{\pi} \lambda^2}, \text{ where } \bar{\pi} = \pi(\bar{x}), \delta = \frac{1 + (1 + \lambda^2) \exp\left\{-\frac{5\lambda^2}{4}\right\}}{1 + \exp\left\{-\frac{\lambda^2}{4}\right\}}$$